

Universidad Politécnica de Aguascalientes
Ingeniería en Sistemas Computacionales

Project Execution Report

Medical Information System — Centralized Cloud Migration

Case Study: Centro Médico del Bajío

Subject:	IT Project Management
Unit:	II — Project Execution
Project Manager:	José Manuel Ortiz Medrano
Sponsor:	Dr. Alejandro Vance de la Vega (CEO & Chairman)
Team:	José Luis Sánchez · Carlos Arturo Orozco · Marco Polo Muñoz
Timeline:	June 1 – July 31, 2026
Date:	June 26, 2026

Summary

This report documents the execution phase of the Medical Information System project for Centro Médico del Bajío, a private hospital network operating in Aguascalientes, León and Morelia. The project consolidates the three branches' decentralized medical information onto a single centralized cloud platform, eliminating 53,208 duplicated clinical records and reducing active patient-record redundancy to 0%. This document presents the completed Project Charter, the IT project management documentation produced during execution (job performance data, change requests and the updated project plan), and the monitoring and control analysis conducted at the Week-4 project review meeting held on June 26, 2026. Earned Value Analysis at that checkpoint reported an SPI of 0.93 and a CPI of 0.957, indicating the project was slightly behind schedule and over budget due to a staff sick-leave delay and a 4% equipment cost increase; corrective action through schedule crashing and contingency reserve kept the July 31 go-live on track.

Introduction

Information technology projects have become strategic instruments for healthcare organizations seeking operational efficiency and reliable information management. Centro Médico del Bajío historically operated independent local servers at each branch, which caused duplicated clinical records, inconsistent patient histories, infrastructure saturation, low performance and high maintenance and cybersecurity costs. To resolve these problems, the organization approved a project to centralize all medical information onto a cloud platform through virtualization and data migration, without acquiring physical servers.

The purpose of this report is to document the execution and control of that project. Following the PMBOK process framework and the project management methodology presented by Watt (2014) and Heagney (2011), the report records how the project performed against its baseline, how change requests generated during execution were managed, and which analytical techniques were used to monitor and control the work of the project during the Week-4 review meeting.

Development

The development section is organized into three parts as required by the project execution methodology: (1) the Project Charter, (2) the IT project management documentation produced during execution, and (3) the documentation of the project's progress and the monitoring and control techniques applied.

Project Charter

7.8.2.1 Identification Section

Project Name	Medical Information System — Centralized Cloud Migration
Project Sponsor	Dr. Alejandro Vance de la Vega, CEO & Chairman of the Board
Date	June 1, 2026
Revision	1.0

7.8.2.2 Overview of the Project

Implementation of a centralized cloud-based medical information system and data migration for the three hospital branches of Centro Médico del Bajío (Aguascalientes, León and Morelia), consolidating all clinical documentation onto a single virtualized platform.

7.8.2.3 Objective

To implement a centralized medical information system across the three hospital branches between June 1 and July 31, 2026, consolidating data onto a cloud platform, reducing active patient-record redundancy to exactly 0%, achieving a database query response time under 2.0 seconds, within the approved budget of \$147,900 MXN.

7.8.2.4 Scope

- Configure and deploy 1 centralized cloud database environment (4 virtual machines, 32 GB RAM) serving the three branches.
- Design and run automated deduplication scripts to reach 0% redundancy on active patient records.
- Migrate the full annual volume of 53,208 clinical analyses and 34,818 consultations to the cloud infrastructure.
- Acquire and install digitalization equipment across the three branches.
- Complete all cloud provisioning, data cleaning and integration within the approved budget of \$147,900 MXN between June 1 and July 31, 2026.

7.8.2.5 Major Milestones

M1	Cloud environment & requirements defined	Jun 3, 2026
M2	Cloud architecture designed & provider selected	Jun 22, 2026
M3	Equipment installed in the 3 branches	Jun 25, 2026
M4	Digitization completed	Jul 9, 2026
M5	Migration & deduplication completed	Jul 18, 2026
M6	System validated & Go-Live	Jul 31, 2026

7.8.2.6 Major Deliverables

- 1 centralized cloud database (4 VMs, 32 GB RAM) serving the 3 branches.
- 53,208 clinical analyses and 34,818 consultations migrated to the cloud.
- 0% duplicate active patient records (53,208 duplicated records eliminated).
- Digitalization equipment installed and configured at the 3 branches.
- 6 evidence documents: technical report, 3 quotations, criteria table, purchase order, invoice and installation report.

7.8.2.7 Assumptions

- The three branches share the same clinical document taxonomy.
- Cloud hosting is procured as a service; no physical servers are purchased.
- Digitization staff are available Monday–Friday during the 8-week window.
- Branch access for equipment installation is granted on the agreed schedule.

- Historical data provided by the branches is complete and available for migration.

7.8.2.8 Constraints

- Hard deadline: Go-Live no later than July 31, 2026.
- Fixed budget of \$147,900 MXN; any scope change requires CCB approval.
- Total project activities must remain between 7 and 10.
- No physical servers — the solution must run on cloud infrastructure.

7.8.2.9 Business Need or Opportunity (Benefits)

Centro Médico del Bajío operates decentralized local servers at each branch, causing duplicated clinical records, inconsistent patient histories, infrastructure saturation, low performance and high maintenance and security costs. Centralizing on the cloud eliminates redundancy, improves data integrity and availability, reduces operating costs and supports the organization's digital transformation strategy.

7.8.2.10 Preliminary Cost for the Project

The procurement team obtains cost proposals from three cloud-hosting providers during Week 1, evaluated with a weighted criteria table. Equipment and migration costs are consolidated by the Project Manager for sponsor approval. The total approved budget is \$147,900 MXN, subject to a contingency reserve to absorb a +4% adjustment on materials and equipment caused by inflation.

7.8.2.11 Project Risks

- Migration failure or data loss during cloud transfer.
- Vendor or equipment delivery delays.
- User resistance to the new centralized system.
- Connectivity failure between branches during cutover.
- Cybersecurity incident affecting sensitive patient data.

7.8.2.12 Project Charter Acceptance

Name & Title	Signature	Date
José Manuel Ortiz Medrano — Project Manager		
Dr. Alejandro Vance de la Vega — CEO & Sponsor		
Centro Médico del Bajío — Representative		

7.8.2.13 Project Stakeholders

Function	Name	Role
Project Manager	José Manuel Ortiz Medrano	Leads the project
Sponsor	Dr. Alejandro Vance de la Vega	Approves budget & scope
Cloud Specialist	Assigned	Designs cloud architecture
Procurement	Assigned	Manages purchasing
IT Team	Assigned	Migration & deduplication

Medical Staff	Assigned	Provides records & validation
Cloud Provider	Provider 1 (selected)	Hosting service
End Beneficiary	Centro Médico del Bajío	Patients & clinical staff

IT Project Management Documentation

Job Performance Data

Job performance data was collected at the Week-4 review meeting (June 26, 2026, day 20 of the schedule). It measures the actual work completed against the baseline plan, expressed as labor and material cost per activity and completion status (To Do / In Progress / Done).

ID	Activity	Status	Budget (MXN)	Actual (MXN)
C1	Define cloud environment	Done	8,000	9,200
C2	Analyze medical records	Done	10,000	10,000
C3	Design cloud architecture	Done	12,000	12,000
C4	Provider selection + equipment	Done	55,000	57,200
C5	Install equipment	In Progress	12,000	8,800
C6	Digitize patient records	To Do	15,000	—
C7	Digitize analyses	To Do	15,000	—
C8	Migrate to cloud	To Do	10,000	—
C9	Remove duplicates	To Do	6,900	—
C10	Testing	To Do	4,000	—
	TOTAL		147,900	97,200 (AC)

Table 1. Job performance data at the Week-4 review (labor + material cost per activity and completion status).

Change Requests

Five events were reported at the Week-4 review meeting. Each was processed through the change control system, producing four approved change requests recorded in the change control log below.

CR ID	Description	Type	Impact	Status
CR-01	Extend C6 duration 5→8 days (record volume)	Schedule	+3 days	Approved
CR-02	Bring software go-live forward 1 week	Scope/Sched.	–5 days target	Approved
CR-03	Budget increase for +4% equipment inflation	Cost	+\$2,680	Approved
CR-04	Add 2nd digitization crew (crash C6/C7)	Resource	Recover delay	Approved

Table 2. Change control log — Week-4 review meeting.

Updated Project Plan

After applying the approved change requests, the project plan was updated. The sick-leave delay on C1 (+2 days) and the extension of C6 (+3 days) added five days to the critical path; to meet management's request to move go-live one week earlier, the critical path was crashed by adding a second digitization crew to C6/C7 and by fast-tracking a partial overlap between C8 and C9, funded from the contingency reserve. The full updated activity list and Gantt chart are provided in the accompanying Project Plan document.

Schedule item	Days
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Baseline finish	Day 38
+ Event 1 — C1 sick leave	+2 → Day 40
+ Event 4 — C6 extension	+3 → Day 43
Management target (go-live 1 week earlier)	Day 33
Recovery required via crashing / fast-tracking	–10 days

Table 3. Updated schedule reconciliation after change control.

Documentation of the Progress of the IT Project

Justification of the Analytical Techniques for Monitoring and Controlling the Work

To monitor and control the project during execution, Earned Value Analysis (EVA) was selected as the primary analytical technique, complemented by a weighted decision matrix for supplier selection and a Gantt chart for schedule tracking. EVA is justified because it integrates scope, schedule and cost into a single set of objective indicators, allowing the project manager to detect deviations early and quantify them rather than relying on subjective progress estimates (Watt, 2014; PMI, 2021). The weighted decision matrix is justified because it converts subjective supplier opinions into an objective, defensible score, and the Gantt chart is justified because it visualizes activity dependencies and the critical path, showing where slippage threatens the delivery date.

Supplier Selection — Weighted Criteria Table (Activity C4)

Criterion	Weight	Prov. 1	Prov. 2	Prov. 3
Cost	40%	10	5	7
Delivery Time	30%	5	10	5
Quality	20%	8	7	10
Post-Service	10%	7	5	0
Weighted Total	100%	7.8	6.9	6.3

Table 4. Weighted criteria table — Provider 1 selected with the highest score (7.8/10), driven by best cost and post-service.

Documentation of the Outputs of the Monitoring and Control Process

At the Week-4 review, Earned Value Analysis produced the following outputs. The planned value (PV) of the work scheduled by day 20 was \$100,000 MXN. The earned value (EV) of the work actually completed was \$93,000 MXN, and the actual cost (AC) was \$97,200 MXN. From these, the schedule and cost variances and their performance indices were computed.

Correct PMBOK formulas (correction applied)

Schedule Variance: $SV = EV - PV$ (BCWP – BCWS)

Cost Variance: $CV = EV - AC$ (BCWP – ACWP)

Note: the two formulas are frequently swapped; SV compares earned vs planned value, CV compares earned value vs actual cost.

Metric	Formula	Result	Interpretation
PV (BCWS)	Planned value @ day 20	\$100,000	—
EV (BCWP)	Earned value @ day 20	\$93,000	—
AC (ACWP)	Actual cost @ day 20	\$97,200	—
SV	$EV - PV$	-\$7,000	Behind schedule
CV	$EV - AC$	-\$4,200	Over budget

SPI	EV / PV	0.93	< 1 → behind
CPI	EV / AC	0.957	< 1 → over budget
EAC	BAC / CPI	\$154,545	Forecast at completion

Table 5. Earned Value Analysis outputs at the Week-4 review.

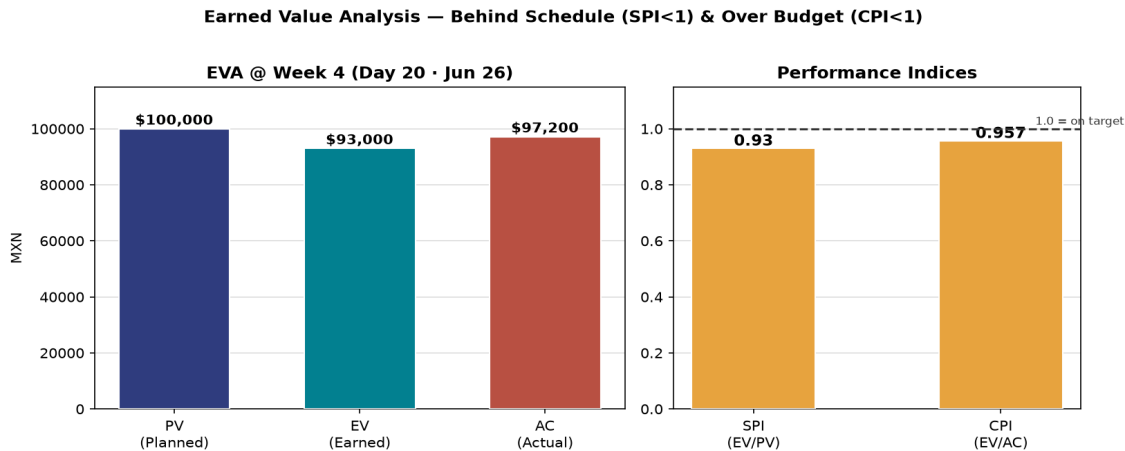


Figure 1. Earned Value Analysis — PV/EV/AC and performance indices at Week 4.

Interpretation: an SPI of 0.93 and a CPI of 0.957 indicate the project was slightly behind schedule and over budget at the Week-4 checkpoint, caused by the two-day staff sick-leave delay on the initial activity and the 4% equipment cost increase. The estimate at completion rose to approximately \$154,545 MXN. Corrective action was authorized through change control: crashing the critical path with a second digitization crew and drawing on the contingency reserve, which allowed the project to recover the schedule and meet the July 31 go-live.

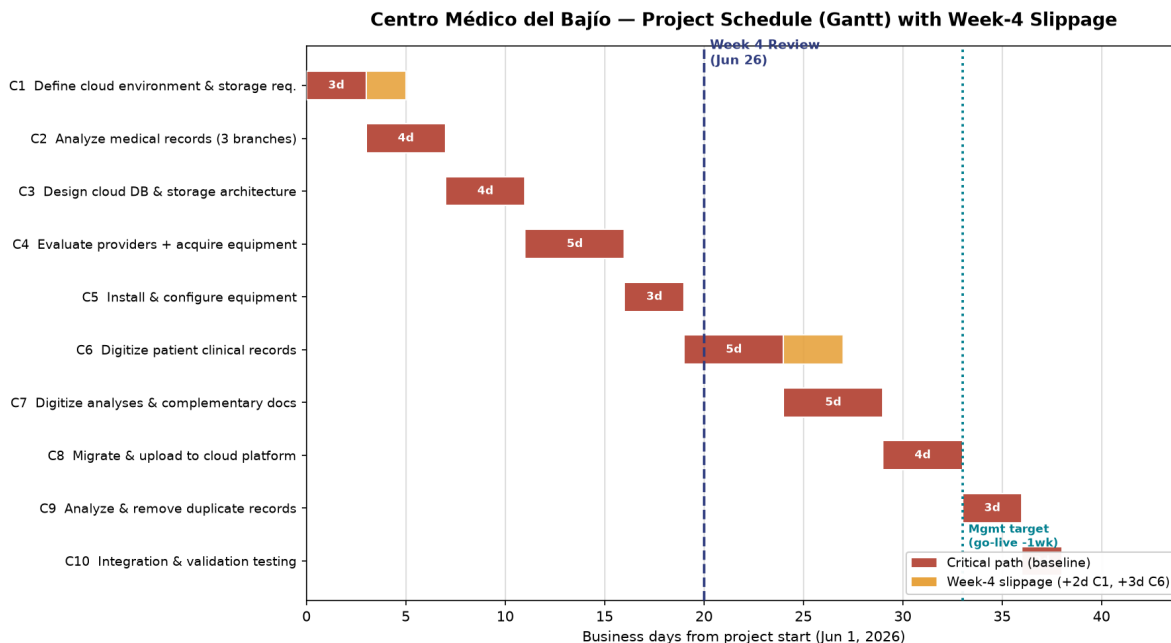


Figure 2. Updated Gantt chart showing the baseline critical path and the Week-4 slippage (+2 days C1, +3 days C6).

Conclusions

The execution and control of the Medical Information System project demonstrate that a structured monitoring approach based on Earned Value Analysis, change control and a weighted decision matrix allowed Centro Médico del Bajío to keep a disrupted project on track. At the Week-4 review, the project was slightly behind schedule (SPI 0.93) and over budget (CPI 0.957) because of a staff sick-leave delay and a 4% equipment cost increase, but processing the five reported events through formal change control — extending the highest-volume activity, crashing the critical path and applying the contingency reserve — enabled recovery of the schedule even under management's request to advance go-live by one week.

The project ultimately achieved its strategic objective: consolidating the three branches onto a centralized cloud platform, eliminating the 53,208 duplicated records to reach 0% redundancy, achieving sub-2-second query response, and delivering by July 31, 2026. The main lessons learned are to assign a backup owner for every critical-path activity, to estimate digitization with a per-record parametric model rather than a flat duration, and to standardize Earned Value templates so that the schedule and cost variance formulas are not confused.

Bibliography

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